

# AML/BSA COMPLIANCE AUTOMATION FRAMEWORK

Digital Assets & VASP | FCF Global Transactions  
FinCEN | OFAC | FATF | BSA | KYC/AML | Travel Rule

|                     |                       |                       |                     |
|---------------------|-----------------------|-----------------------|---------------------|
| 6                   | 40+                   | 18                    | 16                  |
| Framework<br>Layers | Control<br>Categories | Mandatory<br>Controls | AI/ML<br>Components |

A comprehensive six-layer technology and process framework for automating AML/BSA compliance at digital asset startups and VASPs. Covers AI and non-AI controls, end-to-end transaction workflow orchestration, regulatory filing automation, and governance requirements.

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# L1 Regulatory Foundation

## The Legal Perimeter

Map every applicable obligation before a single line of tech is written. Non-compliance here voids everything downstream.

### NON-AI CONTROLS

|                                                                                                                                                                                                               |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>01 FinCEN BSA Program</b><br>Written AML policy, designated BSA Officer, independent audit cycle, ongoing training — all statutorily required for every MSB and VASP.                                      | MANDATORY |
| <b>02 OFAC SDN &amp; Consolidated Sanctions Lists</b><br>Subscribe to OFAC's daily-updated XML feeds. All wallets, counterparties, and beneficial owners screened before onboarding and at every transaction. | MANDATORY |
| <b>03 FATF Travel Rule (31 CFR 1010.410)</b><br>Originator and beneficiary information must travel with every virtual asset transfer >= \$3,000. Implement IVMS101 data standard for interoperability.        | MANDATORY |
| <b>04 State MSB Licensing Matrix</b><br>Map all 50-state licensing requirements. Nevada, California, and New York (BitLicense) have distinct thresholds. Use NMLS for multi-state filings.                    | MANDATORY |
| <b>05 CTR / SAR Filing Thresholds</b><br>CTR: cash transactions >= \$10,000. SAR: suspicious activity >= \$5,000. 30-day filing window for SARs; 15 days for time-sensitive. Retain records 5 years.          | MANDATORY |

### AI / ML COMPONENTS

|                                                                                                                                                            |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>AI-01 Regulatory Change Intelligence</b>                                                                                                                | [ LLM + RAG Pipeline ] |
| NLP agents continuously monitor FinCEN, OFAC, FATF, and SEC rule releases. Auto-flag policy gaps and generate change-impact summaries for the BSA Officer. |                        |
| <b>AI-02 Policy Q&amp;A; Assistant</b>                                                                                                                     | [ Fine-tuned LLM ]     |
| Internal chatbot trained on your AML policy and FinCEN guidance. Compliance staff get instant, citable answers without emailing the BSA Officer.           |                        |

## L2 KYC / CDD / EDD

### Know Your Customer Stack

The onboarding gate. Risk-tier every customer at intake. CDD for standard, EDD for high-risk, Simplified DD for very low-risk retail.

### NON-AI CONTROLS

|                                                                                                                                                                                           |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>01 Identity Document Verification</b><br>Collect government-issued ID. Verify via MRZ parsing, barcode, hologram check.<br>Vendors: Jumio, Onfido, Persona, Stripe Identity.           | HIGH      |
| <b>02 Liveness Detection</b><br>Biometric selfie match against ID photo. 3D liveness prevents spoofing via printed photos or deepfakes. Required for remote onboarding.                   | HIGH      |
| <b>03 Beneficial Ownership (CDD Rule)</b><br>For legal entities: identify all individuals owning >=25% equity OR a single control person. Collect name, DOB, address, ID, TIN.            | MANDATORY |
| <b>04 PEP &amp; Adverse Media Screening</b><br>Screen all customers and UBOs against PEP lists and global adverse media databases. Vendors: Refinitiv World-Check, Dow Jones, LexisNexis. | HIGH      |
| <b>05 Wallet Address Due Diligence</b><br>Verify self-hosted wallet ownership via micro-deposit or signed message proof. Screen deposit/withdrawal addresses via blockchain analytics.    | HIGH      |

### AI / ML COMPONENTS

|                                                                                                                                                                     |                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>AI-01 AI Document Intelligence</b>                                                                                                                               | [ Computer Vision + OCR ]       |
| Vision models extract and validate all fields from ID documents. Cross-reference against authoritative databases. Flag inconsistencies for human review.            |                                 |
| <b>AI-02 Dynamic Risk Scoring Engine</b>                                                                                                                            | [ Gradient Boosting / XGBoost ] |
| ML model assigns a 0-100 risk score at onboarding using 50+ features: nationality, PEP status, source of funds, geography, entity type. Score updates continuously. |                                 |
| <b>AI-03 NLP Adverse Media Mining</b>                                                                                                                               | [ NER + Sentiment NLP ]         |
| Continuously crawl global news in 40+ languages. Entity recognition links articles to customer profiles. Sentiment scoring filters noise from true adverse signals. |                                 |

## L3 Transaction Monitoring

### Real-Time & Retrospective Surveillance

The operational core. Every on-chain and off-chain transaction is screened, scored, and either cleared, flagged, or blocked — in milliseconds.

#### NON-AI CONTROLS

|                                                                                                                                                           |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>01 Blockchain Analytics Platform</b>                                                                                                                   | <b>MANDATORY</b> |
| Integrate Chainalysis Reactor, Elliptic, or TRM Labs. Identify mixer usage, darknet market exposure, sanctions wallet proximity, and ransomware trails.   |                  |
| <b>02 Rule-Based Alert Engine</b>                                                                                                                         | <b>MANDATORY</b> |
| Structuring detection (sub-\$10K deposits), velocity checks, round-dollar amounts, dormant account reactivation, high-risk jurisdiction transfers.        |                  |
| <b>03 OFAC Real-Time Screening</b>                                                                                                                        | <b>MANDATORY</b> |
| Every transaction origin/destination screened against OFAC SDN list before settlement. Hard block on matches. Fuzzy match threshold configurable.         |                  |
| <b>04 Travel Rule Message Broker</b>                                                                                                                      | <b>HIGH</b>      |
| IVMS101 payload transmitted via Notabene, Sygna, or VerifyVASP. Originator data sent on outbound; beneficiary data validated on inbound before crediting. |                  |
| <b>05 Case Management System</b>                                                                                                                          | <b>HIGH</b>      |
| All alerts flow into a case queue (Actimize, Quantexa). Analysts work, document, escalate, or close cases with full audit trail and timestamps.           |                  |

#### AI / ML COMPONENTS

|                                                                                                                                                                                     |                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>AI-01 Behavioral Anomaly Detection</b>                                                                                                                                           | <i>[ Unsupervised ML ]</i>            |
| Autoencoder / isolation forest models each customer's baseline. Flags deviations in amount, frequency, geography, counterparty — catches novel typologies without predefined rules. |                                       |
| <b>AI-02 Graph Neural Network (GNN) Analysis</b>                                                                                                                                    | <i>[ Graph ML ]</i>                   |
| Maps transaction networks as graphs. Detects layering rings, circular fund flows, smurfing networks, and structuring collusion invisible to per-transaction rules.                  |                                       |
| <b>AI-03 SAR Narrative Generator</b>                                                                                                                                                | <i>[ LLM + Structured Prompting ]</i> |
| When analyst decides to file a SAR, AI drafts the FinCEN-formatted narrative using transaction data, customer profile, and alert context. Saves 2-4 hours per SAR.                  |                                       |
| <b>AI-04 False Positive Reducer</b>                                                                                                                                                 | <i>[ Supervised Classification ]</i>  |

Classifier trained on historical analyst decisions scores each new alert with a probability of being a true positive. Prioritizes queue and reduces analyst fatigue.



## L4 On-Chain Intelligence

### Digital Asset-Specific Controls

Traditional AML tools miss the on-chain dimension. This layer is unique to VASPs and digital asset startups — and it is where most gaps exist.

#### NON-AI CONTROLS

|                                                                                                                                                                             |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>01 Multi-Chain Address Labeling</b>                                                                                                                                      | HIGH   |
| Subscribe to labeled address database covering Bitcoin, Ethereum, Solana, Tron, and major L2s. Labels identify exchanges, mixers, darknet markets, and sanctioned entities. |        |
| <b>02 DeFi Protocol Risk Classification</b>                                                                                                                                 | MEDIUM |
| Classify protocols by KYC status, smart contract audit history, and OFAC exposure. Flag transactions involving privacy protocols (Tornado Cash) and non-compliant DEXs.     |        |
| <b>03 NFT &amp; Token Layering Controls</b>                                                                                                                                 | MEDIUM |
| Monitor for NFT round-trip trades, flash loan sequences, and high-frequency DEX swaps that suggest fund movement vs. genuine trading activity.                              |        |
| <b>04 Hot/Cold Wallet Segregation</b>                                                                                                                                       | HIGH   |
| Operational wallets (hot) must be segregated from customer fund wallets. Sweep protocols, signing quorums, and wallet labeling must be documented and auditable.            |        |

#### AI / ML COMPONENTS

|                                                                                                                                                                         |                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>AI-01 Mixer / Tumbler Detection</b>                                                                                                                                  | [ Graph Classification ]  |
| ML classifier identifies CoinJoin clusters, equal-output transactions, peel chains, and chain-hopping sequences even on obfuscated flows where direct links are broken. |                           |
| <b>AI-02 Smart Contract Risk Analyzer</b>                                                                                                                               | [ Static Analysis + LLM ] |
| Static analysis + LLM audit of smart contract code. Flags rug-pull patterns, admin key risks, mint/pause functions, and upgradeability proxies enabling fund theft.     |                           |
| <b>AI-03 Cross-Chain Tracing</b>                                                                                                                                        | [ Multi-chain Graph ML ]  |
| Bridges break the transaction trail. AI agents follow funds across chains by correlating timing, amounts, bridge contract events, and known bridge wallet clusters.     |                           |

## L5 Regulatory Reporting & Filing

### SAR, CTR & Audit Automation

Getting the data right is step one. Filing it correctly, on time, every time — is step two. Automation eliminates the human bottleneck.

#### NON-AI CONTROLS

|                                                                                                                                                                               |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>01 BSA E-Filing System Integration</b>                                                                                                                                     | <b>MANDATORY</b> |
| Direct API or batch submission to FinCEN BSA E-Filing portal. SAR (FinCEN 111) and CTR (FinCEN 112) forms must match current schema versions.                                 |                  |
| <b>02 SAR Decision Audit Trail</b>                                                                                                                                            | <b>MANDATORY</b> |
| Document every alert: who reviewed it, what was found, why filed or not filed. FinCEN examiners require this. Non-documented decisions = regulatory violation.                |                  |
| <b>03 Record Retention System</b>                                                                                                                                             | <b>MANDATORY</b> |
| BSA requires 5-year retention for all transaction records, KYC files, SARs, CTRs, and correspondence. Immutable storage: AWS S3 Object Lock or Azure Immutable Blob.          |                  |
| <b>04 Examiner Readiness Package</b>                                                                                                                                          | <b>HIGH</b>      |
| Pre-built report set for FinCEN/state examinations: risk assessment, policy docs, training records, SAR/CTR filing history, transaction sample, and independent audit report. |                  |

#### AI / ML COMPONENTS

|                                                                                                                                                                     |                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>AI-01 Auto-SAR Drafting Engine</b>                                                                                                                               | <i>[ LLM + Template Engine ]</i>  |
| Pulls alert data, customer profile, transaction timeline, and analyst notes. Structures them into FinCEN-compliant SAR narrative. Dramatically reduces filing time. |                                   |
| <b>AI-02 CTR Auto-Population</b>                                                                                                                                    | <i>[ Rules + LLM QA ]</i>         |
| When cash transaction >= \$10,000 is detected, CTR form fields are auto-populated from customer KYC data and transaction records. Human approves and submits.       |                                   |
| <b>AI-03 Compliance Dashboard Intelligence</b>                                                                                                                      | <i>[ BI + Anomaly Detection ]</i> |
| Real-time KPIs: alert volume, SAR filing rate, false positive ratio, open case age, jurisdictional risk heatmap. AI surfaces anomalies in compliance metrics.       |                                   |



## L6 Governance & Continuous Improvement

### The Oversight Layer

The system must watch itself. Independent audit, model validation, staff training, and a feedback loop that makes the program smarter over time.

#### NON-AI CONTROLS

|                                                                                                                                                                                                                |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>01 Independent Annual AML Audit</b><br>FinCEN requires independent testing of the entire AML program. Engage a qualified third party. Scope: policy, controls, systems, and sampling.                       | MANDATORY |
| <b>02 BSA Officer Accountability</b><br>Designated BSA Officer must have direct board access, authority to halt transactions, and documented escalation paths. Role cannot be outsourced.                      | MANDATORY |
| <b>03 Staff AML Training Program</b><br>Annual training for all employees; role-specific modules for compliance, ops, and customer-facing staff. Maintain completion records and include red flag recognition. | MANDATORY |
| <b>04 Annual Risk Assessment</b><br>Formal written enterprise-wide risk assessment covering customer base, product/service risk, geographic exposure, delivery channels, and emerging typologies.              | MANDATORY |

#### AI / ML COMPONENTS

|                                                                                                                                                                                                                                  |                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>AI-01 Model Risk Management (MRM)</b><br>All AI models in compliance must be validated, documented, version-controlled, and monitored for drift. Track precision/recall metrics. Retrain triggers on performance degradation. | [ MLOps Pipeline ]       |
| <b>AI-02 AI Training Simulations</b><br>Generate synthetic red-flag scenarios and new typology cases. Use them to train staff and validate that AI models correctly flag them. Continuous red-teaming.                           | [ Synthetic Data + LLM ] |
| <b>AI-03 Feedback Loop Engine</b><br>Every analyst decision (file/close) flows back as labeled training data. Models retrained quarterly. Alert rules auto-tuned based on confirmed true-positive patterns.                      | [ Active Learning ML ]   |

# END-TO-END TRANSACTION COMPLIANCE WORKFLOW

How a single transaction flows through the full compliance stack in real time — from customer initiation to FinCEN filing.



## RECOMMENDED TECHNOLOGY STACK

|                                                                                                     |                                                                                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>BLOCKCHAIN ANALYTICS</b><br>Chainalysis · Elliptic · TRM Labs · Merkle Science                   | <b>IDENTITY / KYC</b><br>Persona · Onfido · Jumio · Stripe Identity                           |
| <b>SANCTIONS &amp; PEP SCREENING</b><br>Refinitiv World-Check · Dow Jones Risk · LexisNexis Bridger | <b>TRAVEL RULE PROTOCOL</b><br>Notabene · Sygna Bridge · VerifyVASP · TRISA                   |
| <b>CASE MANAGEMENT</b><br>Actimize · Quantexa · NICE Actimize · ComplyAdvantage                     | <b>AI / ML INFRASTRUCTURE</b><br>AWS SageMaker · Vertex AI · Azure ML · MLflow                |
| <b>GRAPH DATABASE</b><br>Neo4j · Amazon Neptune · TigerGraph                                        | <b>DATA STORAGE &amp; RETENTION</b><br>AWS S3 Object Lock · Azure Immutable Blob · PostgreSQL |
| <b>REGULATORY FILING</b><br>FinCEN BSA E-Filing · NMLS · State portals                              | <b>ORCHESTRATION &amp; MONITORING</b><br>Apache Kafka · Airflow · Datadog · PagerDuty         |

*This framework provides a structured reference for building a FinCEN-compliant AML/BSA program for digital asset startups and VASPs. All AI/ML models used in compliance decisions require Model Risk Management (MRM) documentation, ongoing performance monitoring, and independent validation per regulatory expectations. This document does not constitute legal advice — consult qualified BSA/AML counsel before implementation.*